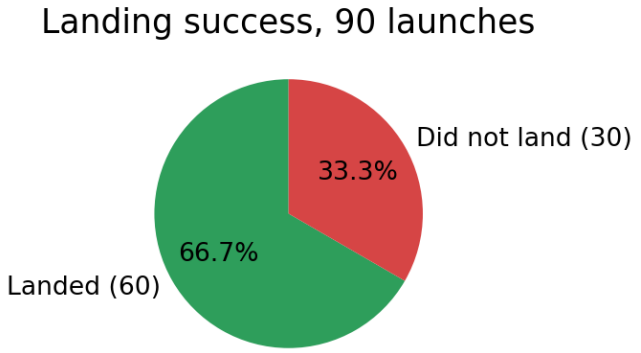
SpaceX Falcon 9 First Stage Landing Prediction

Applied Data Science Capstone

[Your name] | October 2026

Executive Summary

- **Goal:** predict whether a Falcon 9 first stage will land, because reuse is what makes SpaceX launches cheaper (about $62M vs $165M+ from other providers).
- **Data:** 90 Falcon 9 launches, 83 prepared features; SQL analysis on 101 mission records.
- **Methods:** data wrangling and labeling, SQL exploration, interactive Folium maps, and four classification models tuned with 10-fold cross-validation.
- **Key findings:** 66.7% of launches (60 of 90) landed; **KSC LC-39A** has the best success rate (76.9%); sites sit close to coast, rail and highway but about 23 km from the nearest city.
- **Result:** all four models reach **83.3% test accuracy** (15 of 18); the **decision tree** has the best cross-validation accuracy (87.5%).



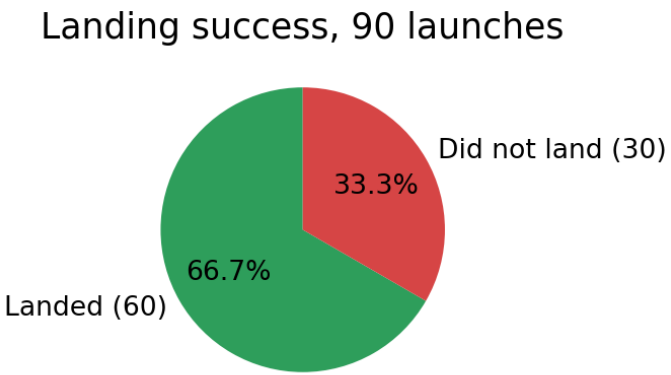
Introduction

- SpaceX advertises Falcon 9 launches at about 62 million dollars; competitors charge 165 million or more. Most of the saving comes from landing and reusing the first stage.
- If we can predict whether the first stage lands, we can estimate the real cost of a launch, useful for an alternative company bidding against SpaceX.
- **Questions we answer:**
- 1. What factors (site, orbit, payload, booster) relate to landing success?
- 2. Does launch-site location matter?
- 3. Which classification model predicts landing success best?

Why it matters: a reliable landing predictor turns launch data into a cost estimate.

Data Collection and Data Wrangling Methodology

- **Collection:** launch records were gathered with the SpaceX REST API and supplemented by web-scraping Falcon 9 launch tables; the result is one dataset of launches, booster, payload, orbit, site and outcome.
- **Wrangling:** missing values were identified (only *LandingPad* had many gaps), numeric and categorical columns were separated, and categorical features (orbit, launch site, landing pad, booster serial) were one-hot encoded and cast to float64 (80 columns in the EDA lab; the ML lab uses the provided 83-column file).
- **Label creation:** the *Outcome* column was converted to a binary **Class**: 0 for failed or unattempted landings (False Ocean, False ASDS, False RTLS, None ASDS, None None), 1 for successful landings.
- **Result:** 60 successes and 30 failures, a 66.7% success rate, which becomes the baseline for the models.

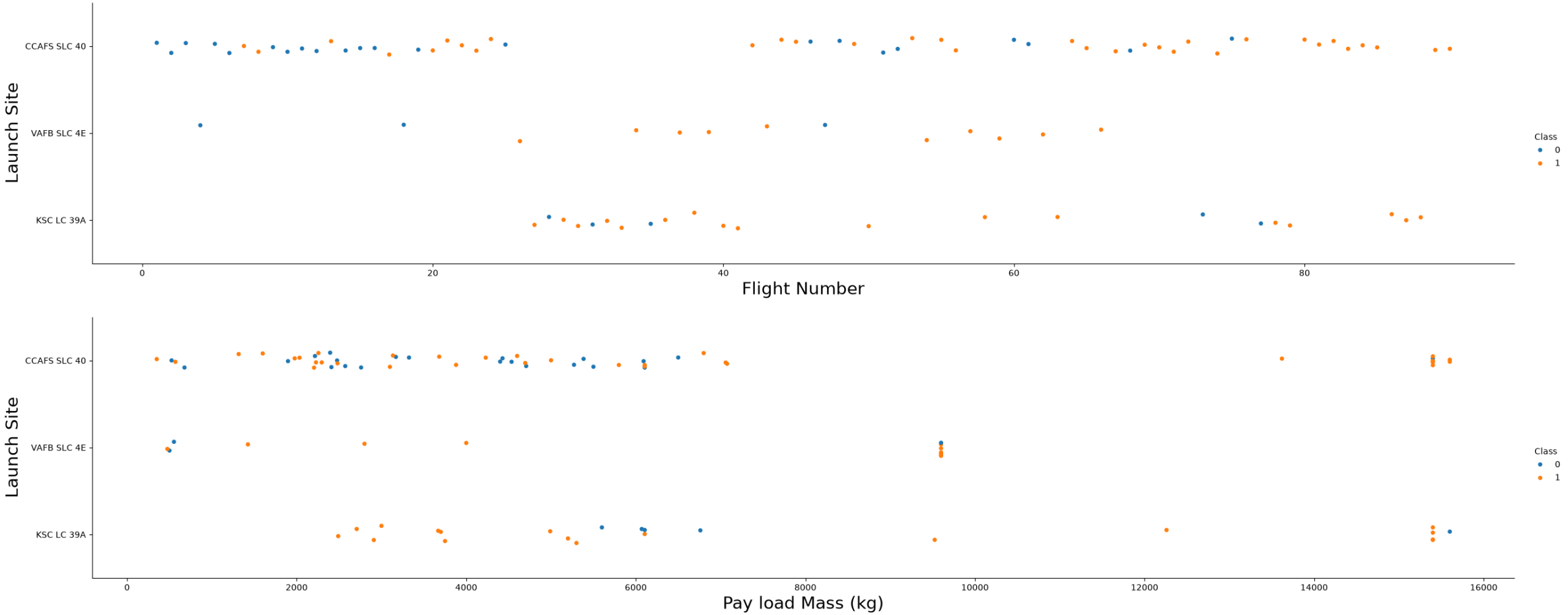


EDA and Data Visualization Methodology

- **Tools:** pandas for grouping and feature engineering, seaborn and matplotlib for charts.
- **Scatter (catplot) charts** with the landing result (Class) as color: Flight Number vs Launch Site, Payload vs Launch Site, Flight Number vs Orbit and Payload vs Orbit. They show how experience, payload and location relate to success.
- **Bar chart** of average Class (success rate) per orbit type, to compare orbits at a glance.
- **Line chart** of the average success rate per year, to see the trend over time.
- **Feature engineering:** selected 12 features, one-hot encoded Orbit, LaunchSite, LandingPad and Serial with *get_dummies*, then cast everything to float64 (90 rows x 80 columns), ready for modelling.

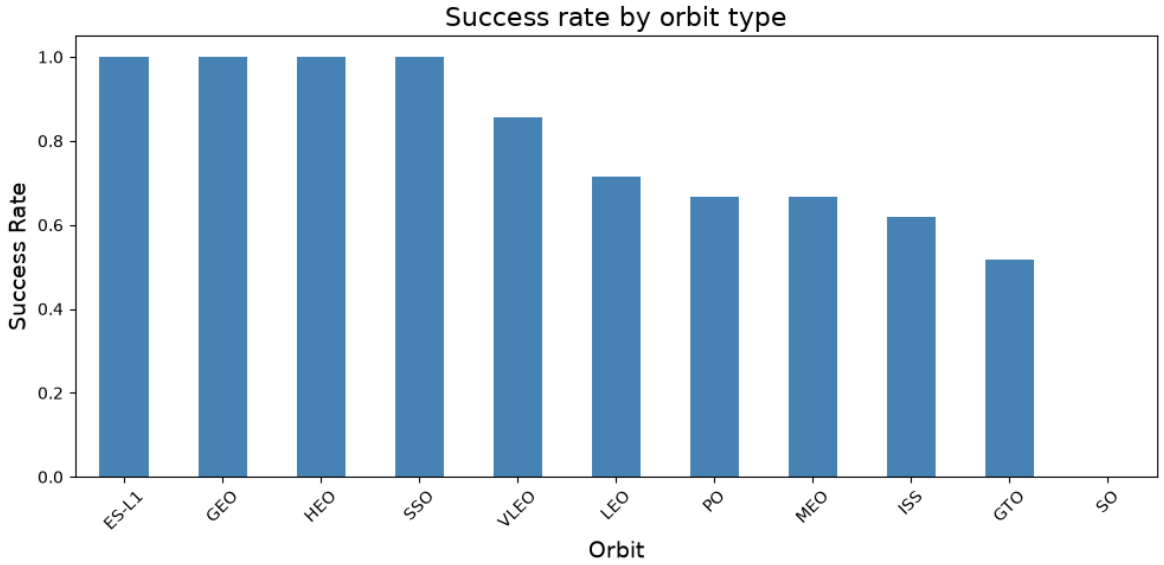
How to read the charts: blue = failed landing (Class 0), orange = successful landing (Class 1).

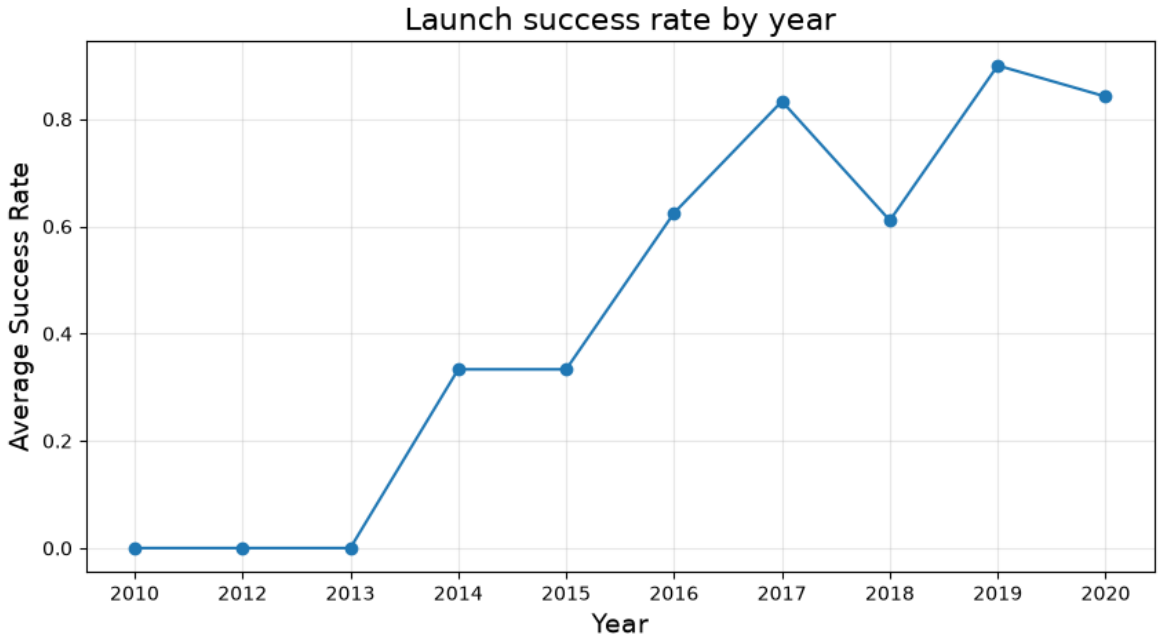
EDA with Visualization (1/3): Launch Sites



Flight Number vs Launch Site: early CCAFS SLC 40 flights mostly failed, later ones mostly landed. KSC LC 39A starts at about flight 26 and mostly succeeds. **Payload vs Launch Site:** VAFB SLC 4E launched nothing above about 10,000 kg.

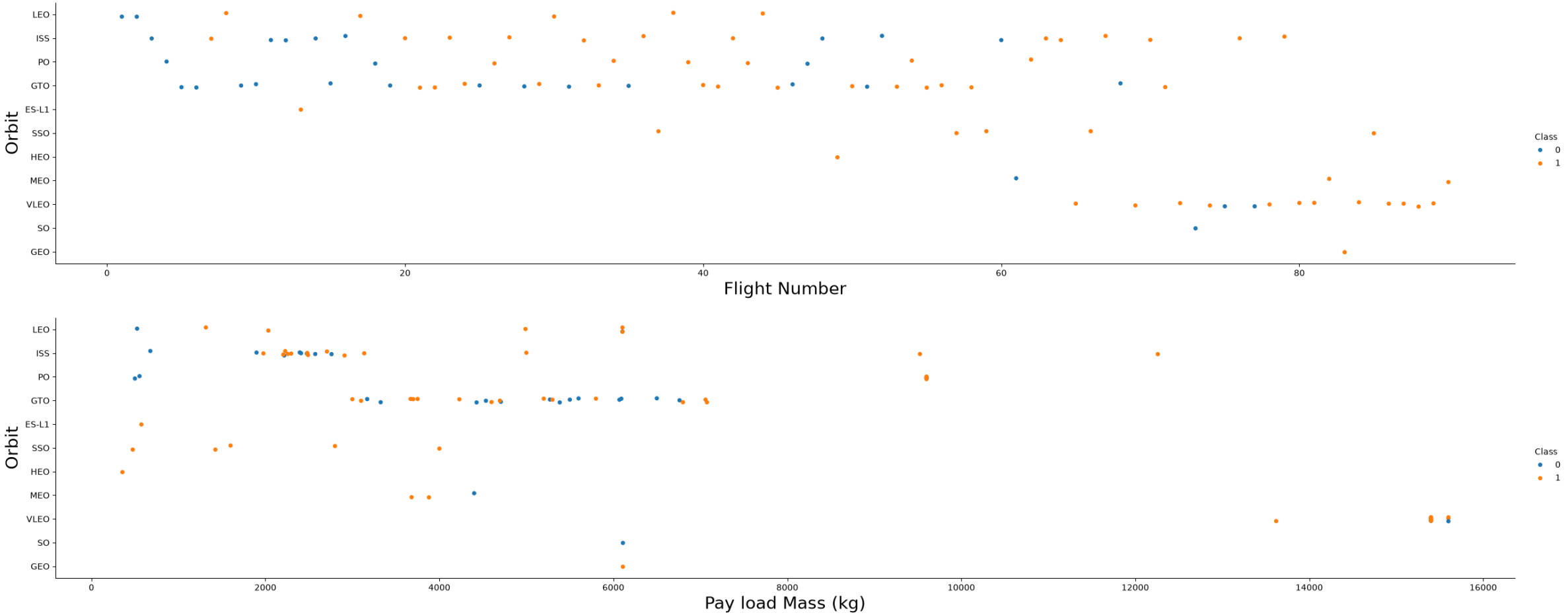
EDA with Visualization (2/3): Orbit and Yearly Trend





- **Orbits:** ES-L1, GEO, HEO and SSO reach 100% (only 1 to 5 launches each); VLEO 85.7%, LEO 71.4%, ISS 61.9%, GTO only 51.9%.
- **Yearly trend:** 0% success from 2010 to 2013, 33% in 2014 and 2015, 62.5% in 2016, 83% in 2017, a dip to 61% in 2018, then 90% in 2019 and 84% in 2020.

EDA with Visualization (3/3): Flight Number and Payload by Orbit



Flight Number vs Orbit: LEO and ISS failures cluster in early flights and later flights mostly land; GTO stays mixed at any flight number; the late VLEO flights almost all land. **Payload vs Orbit:** heavy payloads mostly landed; GTO payloads (about 3,000 to 7,000 kg) show both outcomes.

EDA with SQL (1/4): Launch Sites and Payload

Task	SQL query	Result
1. Unique launch sites	`SELECT DISTINCT Launch_Site FROM SPACEXTABLE;`	4 sites: CCAFS LC-40, VAFB SLC-4E, KSC LC-39A, CCAFS SLC-40
2. Sites starting with CCA	`SELECT * FROM SPACEXTABLE WHERE Launch_Site LIKE 'CCA%' LIMIT 5;`	5 records, all CCAFS LC-40, from 2010-06-04 to 2013-03-01 (first Dragon test flights; payloads of 0, 0, 525, 500 and 677 kg)
3. Total payload, NASA (CRS)	`SELECT SUM(PAYLOAD_MASS__KG_) FROM SPACEXTABLE WHERE Customer = 'NASA (CRS)';`	**45,596 kg**

The earliest launches all came from Cape Canaveral LC-40 and carried small or test payloads. NASA resupply (CRS) missions have lifted over 45 tonnes in total.

EDA with SQL (2/4): Boosters and Landings

Task	SQL query	Result
4. Average payload, F9 v1.1	`SELECT AVG(PAYLOAD_MASS__KG_) FROM SPACEXTABLE WHERE Booster_Version = 'F9 v1.1';`	**2,928.4 kg**
5. First ground pad success	`SELECT MIN(Date) FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (ground pad)';`	**2015-12-22**
6. Drone ship success, payload 4,000 to 6,000 kg	`SELECT DISTINCT Booster_Version FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (drone ship)' AND PAYLOAD_MASS__KG_ > 4000 AND PAYLOAD_MASS__KG_ < 6000;`	F9 FT B1022, F9 FT B1026, F9 FT B1021.2, F9 FT B1031.2

Ground pad landings began in December 2015. Four Full Thrust boosters landed mid-weight payloads on drone ships.

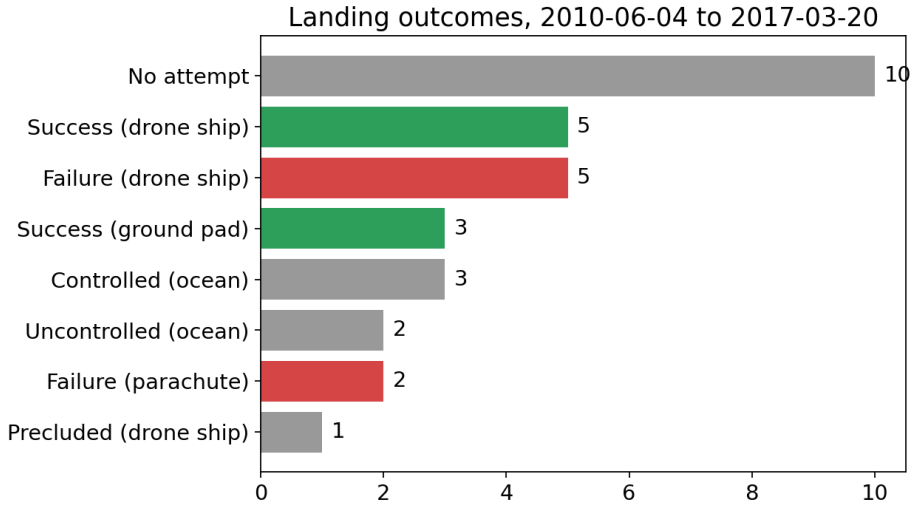
EDA with SQL (3/4): Mission Outcomes and Heaviest Payload

Task	SQL query	Result
7. Mission outcomes	`SELECT Mission_Outcome, COUNT(*) AS Total FROM SPACEXTABLE GROUP BY Mission_Outcome;`	Success: 98 (plus 1 stored with a trailing space) | Success (payload status unclear): 1 | Failure (in flight): 1
8. Boosters with the maximum payload	`SELECT DISTINCT Booster_Version FROM SPACEXTABLE WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTABLE);`	12 boosters, all F9 B5: B1048.4, B1049.4, B1051.3, B1056.4, B1048.5, B1051.4, B1049.5, B1060.2, B1058.3, B1051.6, B1060.3, B1049.7

Mission reliability is very high: 100 successes against 1 in-flight failure. The heaviest payloads were all flown on Block 5 boosters, the final and most reusable Falcon 9 design.

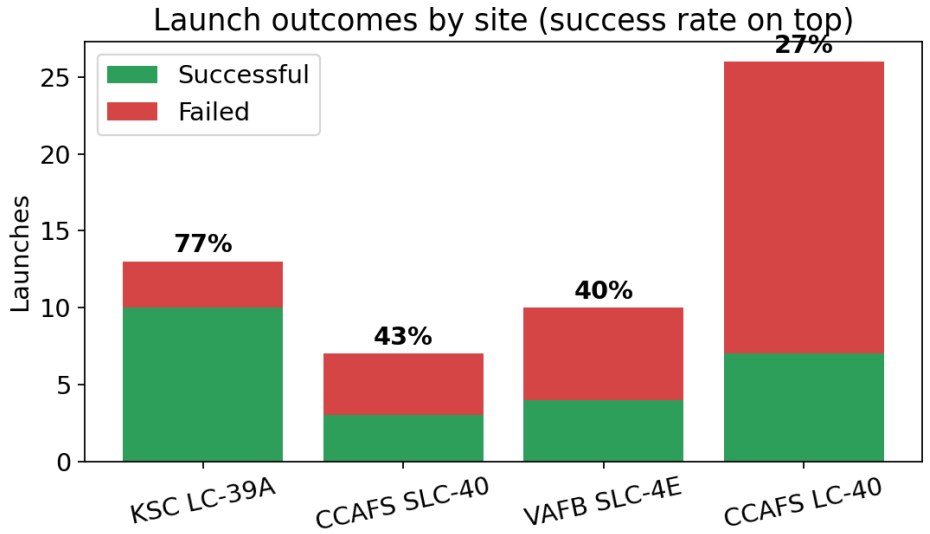
EDA with SQL (4/4): 2015 Failures and Landing Outcome Ranking

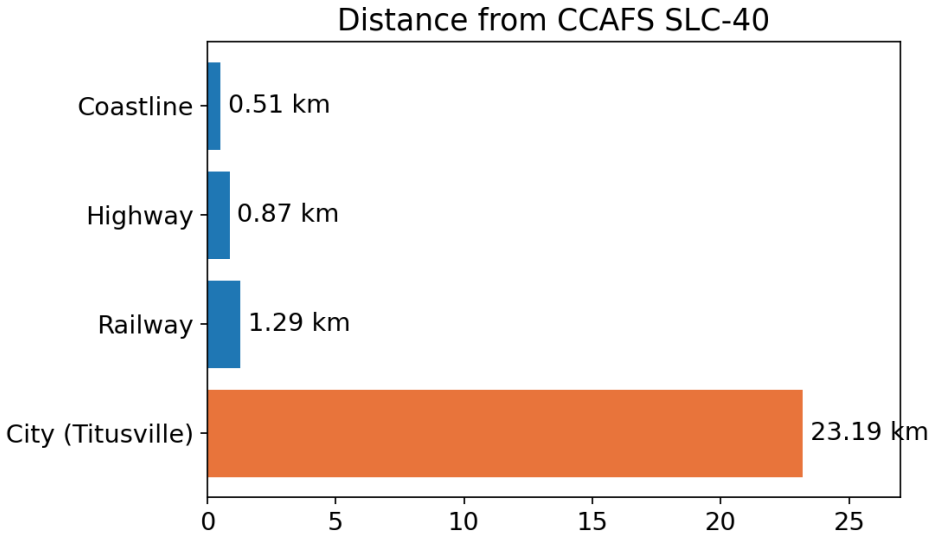
Task	SQL query	Result
9. 2015 drone ship failures	`SELECT substr(Date,6,2) AS Month, Landing_Outcome, Booster_Version, Launch_Site FROM SPACEXTABLE WHERE Landing_Outcome = 'Failure (drone ship)' AND substr(Date,0,5) = '2015';`	2 rows: month 01, booster F9 v1.1 B1012; month 04, booster F9 v1.1 B1015; both from CCAFS LC-40
10. Rank landing outcomes, 2010-06-04 to 2017-03-20	`SELECT Landing_Outcome, COUNT(*) AS Outcome_Count FROM SPACEXTABLE WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY Landing_Outcome ORDER BY Outcome_Count DESC;`	See chart: No attempt 10; drone ship success 5 and failure 5; ground pad success 3



- Both 2015 drone ship failures were early attempts on the F9 v1.1 booster, before SpaceX perfected the technique.
- Through early 2017 drone ship landings split evenly, 5 successes to 5 failures, so landing success improved with experience.

Interactive Map with Folium: Launch Sites





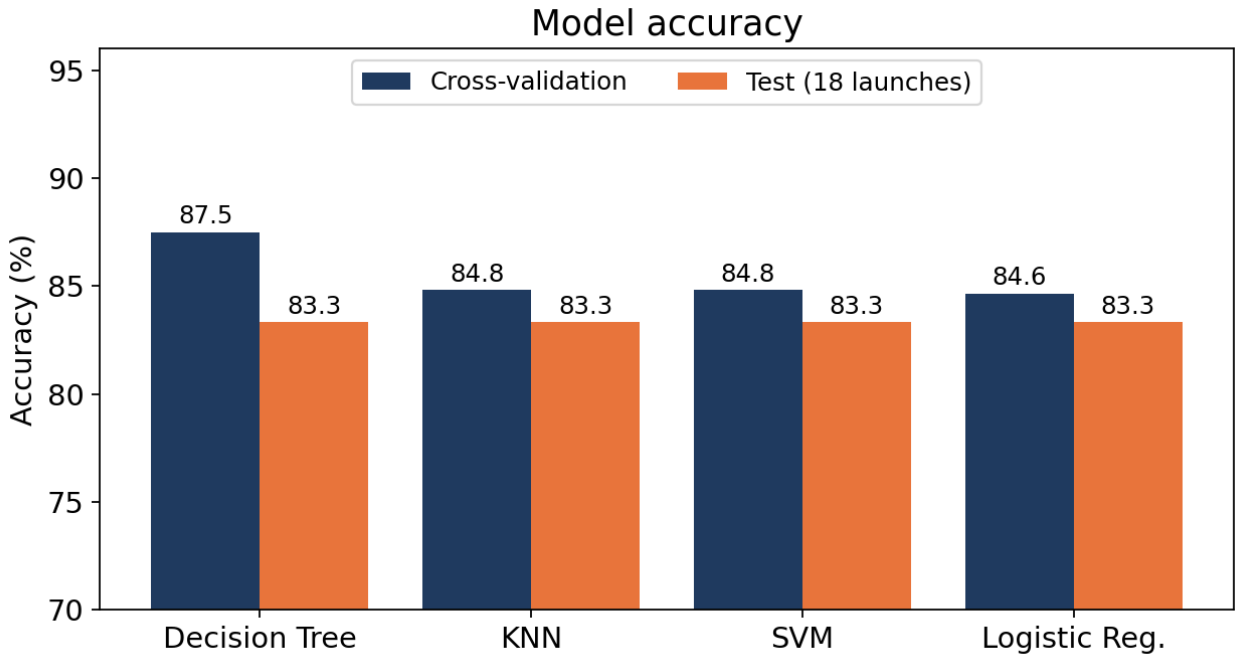
- **Equator:** not close. Sites are at about 28.5 N (Florida) and 34.6 N (California).
- **Coast:** all sites are near the coast, so failed or spent stages fall into the ocean, away from people.
- **Proximities (CCAFS SLC-40):** coastline 0.51 km, highway 0.87 km, railway 1.29 km (supply access); nearest city 23 km (safety distance).
- **Best site:** KSC LC-39A, 10 successes of 13 launches (76.9%).

Predictive Analysis (Classification): Method

- **Features:** 83 prepared columns, standardized with StandardScaler. **Target:** Class (1 = landed).
- **Split:** 80/20 with random_state = 2, giving 72 training and 18 test launches.
- **Models:** logistic regression, support vector machine, decision tree, k-nearest neighbors.
- **Tuning:** GridSearchCV with 10-fold cross-validation on the training set.

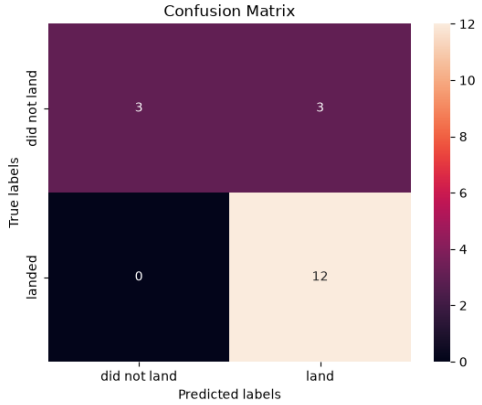
Model	Parameters searched
Logistic regression	C in {0.01, 0.1, 1}; L2 penalty; lbfgs
SVM	kernel: linear, rbf, poly, sigmoid; C and gamma: 5 log-spaced values, 0.001 to 1000
Decision tree	criterion, splitter, max depth 2 to 18, min leaf, min split, max features
KNN	1 to 10 neighbors; 4 algorithms; p = 1 or 2

Predictive Analysis Results



Model	Best parameters	CV acc.	Test acc.
Decision tree	gini, depth 4, leaf 2, split 5, best	87.5%	83.3%
KNN	10 neighbors, auto, p=1	84.8%	83.3%
SVM	sigmoid, C=1, gamma=0.0316	84.8%	83.3%
Logistic reg.	C=0.01, L2, lbfgs	84.6%	83.3%

Best model: decision tree (highest cross-validation accuracy). With only 18 test launches, one launch moves accuracy by 5.6 points, so the models are statistically tied on the test set.



Confusion matrix (identical for all four models): 12 landings correct, 3 non-landings correct, **3 false positives, 0 false negatives**.

Insights and Discussion

- **Every model errs the same way:** none missed a real landing, but each predicted a landing that did not happen 3 times. They are optimistic, which matters when pricing a bid.
- **Location matters:** KSC LC-39A (76.9%) vs CCAFS LC-40 (26.9%). Part of this is time: LC-40 handled many early launches, before landing techniques matured.
- **Learning curve:** the yearly success rate rose from 0% (2010 to 2013) to 83% in 2017 and about 90% in 2019, with one dip to 61% in 2018; orbit also matters, with GTO (51.9%) far harder than SSO (100%).
- **Limits:** 90 launches, 18 test samples, and improvement over time means launches are not independent. A time-based split and more data would be a stricter test.

Conclusion

- Falcon 9 first stage landing success can be predicted with about **83% test accuracy**, against a 66.7% baseline.
- The **decision tree** is recommended (87.5% cross-validation accuracy); all four models tied on the test set.
- **Launch site and experience** relate strongly to success: KSC LC-39A leads at 76.9%.
- Sites are placed near the coast, rail and highway but far from cities (23 km), balancing logistics and safety.
- **Next steps:** add more launches, validate over time, try ensemble models such as random forests, and use predicted landing probability to estimate launch price.